

PSQL COMMANDS :-

PSQL VERSION COMMAND:-

```
odoo@ICAPC0051:~$ psql --version
psql (PostgreSQL) 14.15 (Ubuntu 14.15-0ubuntu0.22.04.1)
```

TO CHECK DATABASE NAME:-

```
odoo@ICAPC0051:~$ psql -l
```

List of databases					
Name	Owner	Encoding	Collate	Ctype	Access privileges
postgres	postgres	UTF8	en_IN	en_IN	
q1	odoo	UTF8	en_IN	en_IN	
task	odoo	UTF8	en_IN	en_IN	
template0	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
test-18	odoo	UTF8	C	en_IN	

(6 rows)

PG CLUSTER COMMAND:-

```
odoo@ICAPC0051:~$ pg_lsclusters
```

Ver	Cluster	Port	Status	Owner	Data directory	Log file
14	main	5432	online	postgres	/var/lib/postgresql/14/main	/var/log/postgresql/postgresql-14-main.log

TO CREATE NEW DATABASE:-

```
odoo@ICAPC0051:~$ createdb demo
odoo@ICAPC0051:~$ psql -l
```

List of databases					
Name	Owner	Encoding	Collate	Ctype	Access privileges
demo	odoo	UTF8	en_IN	en_IN	
postgres	postgres	UTF8	en_IN	en_IN	
q1	odoo	UTF8	en_IN	en_IN	
task	odoo	UTF8	en_IN	en_IN	
template0	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
test-18	odoo	UTF8	C	en_IN	

(7 rows)

TO DROP DATABASE:-

```
odoo@ICAPC0051:~$ dropdb demo
odoo@ICAPC0051:~$ psql -l
```

List of databases					
Name	Owner	Encoding	Collate	Ctype	Access privileges
postgres	postgres	UTF8	en_IN	en_IN	
q1	odoo	UTF8	en_IN	en_IN	
task	odoo	UTF8	en_IN	en_IN	
template0	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	en_IN	en_IN	=c/postgres + postgres=CTc/postgres
test-18	odoo	UTF8	C	en_IN	

(6 rows)

QUE. what is difference between sql and psql list all

SQL	PSQL
It is a database management system.	It is an object-relational database management system
Microsoft develops it.	The PostgreSQL global development group develops it.
It is a commercial tool.	It is an open-source tool.
Language : C++, PHP, C#, Go, Java, Python JavaScript (Node.js), PHP, R, Visual Basic, and Ruby.	Language : C, C++, Java, .Net, Perl, Python, Tcl, JavaScript (Node.js), and PHP
It is compatible with SQL language.	It support various programming languages.

It is highly scalable.	It support multiple feature of SQL.
SQL is not case sensitive.	PostgreSQL is case sensitive.
It is a platform independent.	It is free to download.
SQL is a Structure query language.	Psql is a advance tool.

QUE. create/delete multiple database.

- Create database: createdb demo
Delete database: dropdb demo
- Create database: createdb demo1
Delete database: dropdb demo1
- Create database: createdb demo2
Delete database: dropdb demo2

QUE. check use of alter command for tables

```
task=# ALTER TABLE employees ADD Salary INT;
ALTER TABLE
task=# SELECT * FROM employees;
 employee_id | name  | department | salary
-----+-----+-----+-----
          101 | Alice | ERP        |
          102 | BOB  | SALES      |
          103 | Charli | PHP       |
(3 rows)
```

```
task=# ALTER TABLE employees DROP Salary;
ALTER TABLE
task=# SELECT * FROM employees;
 employee_id | name | department
-----+-----+-----
          101 | Alice | ERP
          102 | BOB   | SALES
          103 | Charli | PHP
(3 rows)
```

For adding a new column : ALTER TABLE employees
ADD COLUMN salary BOOLEAN;

For dropping a column : ALTER TABLE employees
DROP COLUMN salary;

For renaming a column : ALTER TABLE employees
RENAME COLUMN name TO e_name;

####TASK####

####CREATING employees TABLE####

CREATE TABLE employees(employee_id SERIAL PRIMARY KEY, name VARCHAR(100),
department VARCHAR(100));

####INSERTING DATA INTO employees TABLE####

INSERT INTO employees(employee_id,name, department) values(101, 'Alice', 'ERP'),(102,
'Bob', 'SALES'),(103, 'Charlie', 'PHP');

####TO SHOW ALL DATA FROM EMPLOYEES TABLE####

SELECT * FROM employees;

```
task=# SELECT * FROM employees;
 employee_id | name | department
-----+-----+-----
          101 | Alice | ERP
          102 | BOB   | SALES
          103 | Charli | PHP
```

####CREATING project TABLE####

```
CREATE TABLE project (project_id SERIAL PRIMARY KEY, project_name VARCHAR(100)
NOT NULL, employee_id INT, FOREIGN KEY (employee_id) REFERENCES
employees(employee_id) );
```

####INSERTING DATA INTO project TABLE####

```
INSERT INTO project (project_id, project_name) VALUES (101, 'Web redesign');
INSERT INTO project (project_id, project_name) VALUES (102, 'Ad campaign');
INSERT INTO project (project_id, project_name) VALUES (103, 'Recruitment drive');
```

```
task=# INSERT INTO project (project_id, project_name) VALUES (101, 'Web redesign');
INSERT 0 1
task=# INSERT INTO project (project_id, project_name) VALUES (102, 'Ad campaign');
INSERT 0 1
task=# INSERT INTO project (project_id, project_name) VALUES (103, 'Recruitment drive');
INSERT 0 1
task=# SELECT * FROM project;
 project_id | project_name | employee_id
-----+-----+-----
        101 | Web redesign |
        102 | Ad campaign  |
        103 | Recruitment drive |
(3 rows)
```

####TO SHOW ALL DATA FROM PROJECT TABLE####

```
SELECT * FROM project;
```

JOIN QUERY

```
SELECT e.name, p.project_name
FROM employees e
JOIN project p ON e.employee_id = p.project_id;
```

```
task=# SELECT e.name, p.project_name
FROM employees e
JOIN project p ON e.employee_id = p.project_id;
 name | project_name
-----+-----
 Alice | Web redesign
 BOB   | Ad campaign
 Charli | Recruitment drive
(3 rows)
```